



Commutative, distributive and associative laws

Teacher resource

Question 1

Which one of the following does **not** give the same result as $45 + 45 + 45$?

- ☐ $45 + (2 \times 45)$
- ☐ $(1 \times 45) + (2 \times 45)$
- ☐ $(3 + 5)(40 + 3)$
- ☐ $3(40 + 5)$

Answer: C

Skill: Students apply the commutative, associative and distributive laws to numerical expressions with whole numbers.

Additional questions

1. Use an array to show that $3 \times 6 + 4 \times 3$ gives the same result as 3×10 .
2. Use an array to show that 12×18 can be thought of as $(10 \times 10) + (10 \times 8) + (2 \times 10) + (2 \times 8)$. Use this partitioning to evaluate 12×18 .

Question 2

Which one of the following does **not** give the same result as 8×3.98 ?

- ☐ $8 \times 3 + 8 \times 0.98$
- ☐ $8 \times 4 - 0.02$
- ☐ $8 \times 3 + 8 \times 0.9 + 8 \times 0.08$
- ☐ $8 \times 3 + 8(0.29 + 0.69)$

Answer: B

Skill: Students apply the commutative, associative and distributive laws to numerical expressions with decimals.

Additional questions

Evaluate the following expressions:

1. $0.8 + 0.5 + 0.2 + 0.3$
2. $0.13 + 1.34 + 0.07$
3. $5.4 + 2.8 - 3.4 + 1.2$

Question 3

Which one of the following is **not** equivalent to $a + b + a + b$?

- ☐ $(a + b) + (b + a)$
- ☐ $2(a + b)$
- ☐ $a + a + b + b$
- ☐ $(a + b)(a + b)$

Answer: D

Skill: Students apply the commutative, associative and distributive laws to numerical expressions with variables.

Additional questions

Use an array to represent:

1. $2(a + b)$
2. $a(a + b)$
3. $(a + b)(a + b)$

Question 4

Which one of the following is equivalent to $-(a + 3)$?

- ☐ $-a - 3$
- ☐ $a - 3$
- ☐ $-a + 3$
- ☐ $a + 3$

Answer: A

Skill: Students apply the distributive law to numerical expressions with variables.

Additional questions

1. What is the value of $34 - (17 - 2)$?
2. Evaluate $-4 - (-4 - 4)$

Student activity: Commutative, distributive and associative laws

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- ☐ $(1 \times 45) + (2 \times 45)$
- ☐ $(3 + 5)(40 + 3)$
- ☐ $3(40 + 5)$

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- ☐ $8 \times 3 + 8 \times 0.98$
- ☐ $8 \times 4 - 0.02$
- ☐ $8 \times 3 + 8 \times 0.9 + 8 \times 0.08$
- ☐ $8 \times 3 + 8(0.29 + 0.69)$

Question 4

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- ☐ $-a - 3$
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